

Recall from last class:

- Let A be an $m \times n$ matrix, then

$$A = [\vec{a}_1 \ \dots \ \vec{a}_n], \quad \vec{a}_j \in \mathbb{R}^m, \ 1 \leq j \leq n.$$

If $x \in \mathbb{R}^n$, then we define

$$A\vec{x} = \vec{a}_1x_1 + \vec{a}_2x_2 + \dots + \vec{a}_nx_n.$$

Thus the matrix equation $A\vec{x} = \vec{b}$ is the same as the vector equation

$$\vec{a}_1x_1 + \vec{a}_2x_2 + \dots + \vec{a}_nx_n = \vec{b}.$$

- $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b} \in \text{Span}(\vec{a}_1, \dots, \vec{a}_n)$ where $\vec{a}_j$ is the j -th column of A .
- A set of vectors $\vec{v}_1, \dots, \vec{v}_p$ in $\mathbb{R}^m$ **spans** $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $\vec{v}_1, \dots, \vec{v}_p$.
- **Theorem (Theorem 4 in the book)** Let A be an $m \times n$ matrix, then the following are equivalent.
 1. For each $\vec{b} \in \mathbb{R}^m$, the matrix equation $A\vec{x} = \vec{b}$ is consistent.
 2. Each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A .
 3. The columns of A span $\mathbb{R}^m$.
 4. A has a pivot position in every row.
- **Fact (or Theorem 5 from the book):** If A is a $m \times n$ matrix and $\vec{u}, \vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$, then
 1. $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$
 2. $A(c\vec{u}) = c(A\vec{u})$.

Definition 1. A linear system is called **homogeneous** if it can be written as $A\vec{x} = \vec{0}$ where A is an $m \times n$ matrix, $\vec{x} \in \mathbb{R}^n$, and $\vec{0}$ is the vector of m zeros.

1. Let $A = \begin{bmatrix} 2 & -5 & 8 \\ -2 & -7 & 1 \\ 4 & 2 & 7 \end{bmatrix}$. Find all solutions to $A\vec{x} = \vec{0}$.

Theorem: If $A\vec{x} = \vec{b}$ is consistent for a given $\vec{b}$, and $\vec{p}$ is a solution, then the solution set of $A\vec{x} = \vec{b}$ is the set of vectors of the form $\vec{p} + \vec{v}_h$ where $\vec{v}_h$ is a solution of $A\vec{x} = \vec{0}$.

2. Prove the Theorem.

3. Let $A = \begin{bmatrix} 1 & 0 & 5 \\ -2 & 1 & -6 \\ 0 & 2 & 8 \end{bmatrix}$.

(a) Find all solutions to $A\vec{x} = \vec{0}$.

(b) Let $\vec{b} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$. Describe all solutions to $A\vec{x} = \vec{b}$.

4. Construct a 2 by 2 matrix A such that the solution set to $A\vec{x} = \vec{0}$ is the line in $\mathbb{R}^2$ going through $(0, 0)$ and $(4, 1)$.

Definition 2. A set of vectors $\{\vec{v}_1, \dots, \vec{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \dots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution. The set is called **linearly dependent** if the equation has non-trivial solutions and the equation is called the **dependence relation**.

5. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Is $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ linearly independent?

Let $A = [\vec{a}_1 \dots \vec{a}_n]$, then $\{\vec{a}_1 \dots \vec{a}_n\}$ are linearly independent if and only if $A\vec{x} = \vec{0}$ has only the trivial solution.

6. Is a set with only one vector always linearly independent?

7. If $\vec{0} \in \{\vec{v}_1, \dots, \vec{v}_p\}$, then is the set linearly dependent?

8. If $\vec{0} \in \text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$, then the vectors are linearly independent?

9. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$. Are the vectors linearly independent?

Theorem 7: A set $S = \{\vec{v}_1, \dots, \vec{v}_p\}$ of two or more vectors is linearly dependent if and only if one of the vectors of S is a linearly combination of the others.

Moreover if $\vec{v}_1 \neq \vec{0}$, then some $\vec{v}_j$ for $j > 1$ is a linear combination of $\vec{v}_1, \dots, \vec{v}_{j-1}$.

10. Prove Theorem 7.

11. Let $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}$

(a) Describe $\text{Span}(\vec{u}, \vec{v})$

(b) Prove $\vec{w} \in \text{Span}(\vec{u}, \vec{v})$ if and only if $\{\vec{u}, \vec{v}, \vec{w}\}$ is linearly dependent.

Theorem 8: Any set $\{\vec{v}_1, \dots, \vec{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

12. Prove the theorem.